

Let $p_n = (x_n, y_n)$. Since $(x_n) \subset [a, b]$, and $[a, b]$ is closed and bounded hence sequentially compact, there exists a subsequence (x_{n_k}) such that $x_{n_k} \rightarrow x$ for some $x \in [a, b]$. Now, take the subsequence of (y_n) , $(y_{n_k}) \subset [A, B]$. Since $[A, B]$ is also closed and bounded, it is also sequentially compact, so there exists a subsequence $(y_{n_{k_j}}) \subset [A, B]$ such that $y_{n_{k_j}} \rightarrow y$ for some $y \in [A, B]$, and, since $x_{n_{k_j}}$ is a subsequence of x_{n_k} , it still converges to x . Now, by part (a), we have that $(x_{n_{k_j}}, y_{n_{k_j}}) \rightarrow (x, y) \in [A, B]$, so Q is sequentially compact as required.